

HAOYANG YU

LOW-DIMENSIONAL TOPOLOGY · FLOER HOMOLOGY

yuhyumath@gmail.com

School of Mathematical Sciences, Peking University

EDUCATION

Peking University

2023 – 2027 (expected)

B.S. in Mathematics and Applied Mathematics · Beijing, China

- Overall credit-weighted average **90.17/100** (126 graded credits).
- Mathematics coursework credit-weighted average **93.06/100** (91 graded credits).

RESEARCH INTERESTS

Low-dimensional topology; knot theory; 3-manifold topology; Heegaard Floer homology; monopole Floer homology. Currently exploring mapping class groups and symplectic Floer theory.

RESEARCH EXPERIENCE

The University of Chicago Mathematics REU

Jun – Aug 2026

Participant · Chicago, Illinois

- Participating in an eight-week program of intensive mathematical study and research; preparing a final expository or research paper on a topic of choice.

3-Manifolds and Floer Homology

Sep 2025 – Present

Undergraduate Research Project · School of Mathematical Sciences, Peking University

- Studying Morse homology, Heegaard Floer homology, monopole Floer homology and 3-manifold topology under the supervision of Prof. Jiajun Wang.

SELECTED PROJECTS

Interactive Knot-Theory Visualizations

2026

Independent browser-based software projects

- **Knot Isotopy Explorer.** A visualization of PL knots that constrains vertex motion with Δ -move intersection tests, redraws planar diagrams in real time, detects Reidemeister moves, and computes the Alexander polynomial from a Wirtinger presentation via Fox calculus.
- **Lifting a Knot from Its Diagram.** A tool that reconstructs an embedded spatial PL knot from a planar shadow and over/under data with exact reprojection; for at most 12 crossings it enumerates all crossing assignments and groups them by Alexander polynomial.

HONORS AND AWARDS

2026	National Scholarship, Peking University (Undergraduate)
2026	8th Place, Geometry and Topology Written Examination, 17th S.-T. Yau College Student Mathematics Contest
2025	Honorable Mention, Written Examination in Geometry and Topology, 16th S.-T. Yau College Student Mathematics Contest
2025	First Prize, 36th Beijing College Students' Mathematics Competition (Mathematics Major, Group A)
2025	Merit Student, Peking University (2024–2025 academic year)
2025	Qin Wanshun–Jin Yunhui Scholarship, Peking University (2024–2025 academic year)
2025	Young Talent Support Program in Basic Mathematics, School of Mathematical Sciences, Peking University (supported by Xiwang)
2021, 2022	Gold Medal (First Prize), Chinese Mathematical Olympiad (CMO)

SELECTED MATHEMATICS COURSEWORK

COURSE	GRADE	COURSE	GRADE
Abstract Algebra	93	Homology Theory and Characteristic Classes	99
Advanced Algebra (II)	96	Hyperbolic Partial Differential Equations	A
Complex Geometry	A+	Introduction to Riemannian Geometry	95
Differential Geometry	92	Mathematical Analysis (II)	96
Differential Manifolds and Topology (Honor)	92	Mathematical Analysis (III)	97.5
Dynamical Systems	95	Mathematical Modeling	100
Functional Analysis	93	Ordinary Differential Equations	92
Functions of Real Variables	96	Partial Differential Equations	96
Geometric Analysis	93	Riemann Surfaces	A–
Geometric Group Theory	A+	Symplectic Geometry	92
Geometry I (Honor)	91	Theory of Functions of Complex Variables	99
Groups and Representations	98	Topology	A

Course titles as they appear on the official Peking University transcript. Percentage averages convert the letter grades A+, A and A– to 98, 95 and 91.